Ankur Barsode

Publications

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Updated: 16 September 2026

Refereed Journal Articles

1. **Barsode, Ankur**. "Tool for Accurate Model Selection Using Bayes Factors." *Physical Review D*, vol. 114, 2026, article 064045. [link]

2. **Barsode, Ankur**, Koustav N. Maity, and Parameswaran Ajith. "Search for Strong Lensing of Gravitational Waves in the Binary Black Hole Events from O1–O4a." *The Astrophysical Journal*, vol. 1009, no. 1, 2026, article 43. [link]

3. **Barsode, Ankur**, Koustav N. Maity, and Parameswaran Ajith. "Lensing, Not Luck! Detection Prospects of Strongly Lensed Gravitational Waves." *The Astrophysical Journal*, vol. 998, no. 2, 2026, article 294. [link]

4. **Barsode, Ankur**, Srashti Goyal, and Parameswaran Ajith. "Fast and Efficient Bayesian Method to Search for Strongly Lensed Gravitational Waves." *The Astrophysical Journal*, vol. 980, no. 2, 2025, article 258. [link]

5. **Barsode, Ankur**, and Tirthankar Roy Choudhury. "Efficient Hybrid Technique for Generating Sub-Grid Haloes in Reionization Simulations." *Journal of Cosmology and Astroparticle Physics*, no. 11, 2024, article 036. [link]

6. **Barsode, Ankur**, S. J. Kapadia, and Parameswaran Ajith. "Constraints on Compact Dark Matter from the Nonobservation of Gravitational-Wave Strong Lensing." *The Astrophysical Journal*, vol. 975, no. 1, 2024, article 48. [link]

7. **Barsode, Ankur**, and Aneesh A. M. "Bypassing Traditional Molecular Dynamics with Artificial Neural Networks." *AIP Conference Proceedings*, 2023. [link]

8. Maity, Koustav N., Souvik Jana, Tejaswi Venumadhav, **Ankur Barsode**, and Parameswaran Ajith. "Strong Lensing Cosmography Using Binary-Black-Hole Mergers: Prospects for the Near Future." *Physical Review D*, vol. 113, no. 10, 2026, article 103522. [link]

9. Wright, Mick, et al. "LensingFlow: An Automated Workflow for Gravitational Wave Lensing Analyses." *RAS Techniques and Instruments*, vol. 5, 2026, article rzag003. [link]

Submitted / Under Review

1. Maity, Koustav N., Souvik Jana, **Ankur Barsode**, Tejaswi Venumadhav, and Parameswaran Ajith. "Probing Dark Matter with Strongly Lensed Binary Black Hole Mergers: Prospects in the Near Future." *arXiv*, arXiv:2609.05020, 2026. [link]

2. The LIGO Scientific Collaboration, the Virgo Collaboration, and the KAGRA Collaboration. "GWTC-4.0: Searches for Gravitational-Wave Lensing Signatures." *arXiv*, arXiv:2512.16347, 2025. [link]